

Day 18 - Lecture 18.6: Format Strings (fmt)

1. Overview & Syntax

In Matplotlib, a **Format String** (`fmt`) provides an ultra-compact shorthand to specify **marker**, **line style**, and **color** in a single string argument:

`fmt = '[marker][linestyle][color]'` or `'[color][marker][linestyle]'`

Examples:

- `'o-b'` : Circle marker (`o`), Solid line (`-`), Blue color (`b`)
- `'>--g'` : Triangle-right marker (`>`), Dashed line (`--`), Green color (`g`)
- `'r:'` : Red color (`r`), Dotted line (`:`), No marker
- `'ks'` : Black color (`k`), Square marker (`s`), No line

2. Format String Components Reference Table

Marker Codes	Line Styles	Color Codes
<code>.</code> (point)	<code>-</code> (solid)	<code>b</code> (blue)
<code>o</code> (circle)	<code>--</code> (dashed)	<code>g</code> (green)
<code>s</code> (square)	<code>-. </code> (dash-dot)	<code>r</code> (red)
<code>^</code> (triangle up)	<code>:</code> (dotted)	<code>c</code> (cyan)
<code>></code> (triangle right)	<i>(omitted = no line)</i>	<code>m</code> (magenta)
<code>x</code> (cross) / <code>+</code> (plus)		<code>y</code> (yellow)
<code>*</code> (star) / <code>d</code> (diamond)		<code>k</code> (black) / <code>w</code> (white)

3. Shorthand vs. Explicit Keyword Arguments

Format strings are quick, but explicit keyword arguments offer granular control:

```
# Shorthand (Format String):  
plt.plot(x, y, "o-b", label="Winners")  
  
# Equivalent Explicit Keyword Arguments:  
plt.plot(x, y, color="blue", linestyle="-", marker="o", linewidth=2, markersize=8)
```

4. Code Implementation

```
import matplotlib.pyplot as plt  
  
years = [2008, 2009, 2010, 2011, 2012]  
oscar_revenue = [1005, 170, 427, 133, 232]  
non_oscar_revenue = [378, 2788, 829, 185, 275]  
  
plt.figure(figsize=(9, 5))  
  
# 1. Shorthand Format String  
plt.plot(years, oscar_revenue, "o-b", label="Oscar (fmt: 'o-b')")  
plt.plot(years, non_oscar_revenue, ">--g", label="Non-Oscar (fmt: '>--g')")  
  
# 2. Detailed explicit styling  
plt.plot(years, [r * 0.5 for r in oscar_revenue], color="black", linestyle=":", n  
          linewidth=2, markersize=6, label="Half Revenue (Explicit kwargs)")  
  
plt.title("Format Strings Demonstration", fontsize=14, fontweight='bold')  
plt.xlabel("Years")  
plt.ylabel("Revenue ($M)")  
plt.legend()  
plt.grid(True)  
plt.show()
```

5. Key Takeaways

- Use format strings for rapid prototyping and notebooks.

- Use explicit keyword arguments (`color` , `linestyle` , `linewidth` , `marker`) in production code for readability and maintainability.